

Identity Falsification and Quotas

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Abstract

This paper studies a screening problem in which a manager acquires costly information about candidates who can costlessly misreport their group identity. Group identity is valuable because it sharpens the manager's prior about individual quality, reducing the cost of learning whether a candidate is a good match. Without commitment, informative communication breaks down: in every equilibrium, the manager applies the same hiring rule regardless of reported identity. A quota can sustain truthful reporting. The welfare consequences depend on the information-cost structure. The key property is Prior Invariance: if quotas can compensate for changes in priors while keeping the same state-contingent hiring rule optimal, then truthful reporting can be restored without changing candidates' hiring probabilities. Entropy costs satisfy Prior Invariance, so quotas can improve efficiency by lowering only information costs. Under quadratic costs, by contrast, truthful revelation generally requires a change in the allocation and therefore introduces redistribution.

Keywords: rational inattention, cheap talk, quotas, identity falsification, mechanism design

JEL Classification: D81, D82, D83, J71

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1 Introduction

When individual quality is not directly observable, decision-makers rely on group-level base rates to form priors and then treat otherwise similar candidates differently based on group membership (Phelps, 1972; Arrow, 1973). As a result, candidates whose identities are treated less favorably have an incentive to misreport them. This incentive is not merely theoretical. For example, individuals misreport self-reported race, ethnicity, or caste in response to policy incentives (Antman and Duncan, 2015; Intelligent.com, 2021; Cassan, 2015), and job candidates falsify their educational backgrounds (Attewell and Domina, 2011; ResumeLab, 2023). Whenever a reported characteristic affects how a decision-maker evaluates an applicant, applicants have an incentive to manipulate that report.

At the same time, those same characteristics can be genuinely useful for screening. Group identity can help a decision-maker form a prior about quality, and it can also shape how intensively or in what way a candidate is screened.¹ Bartoš et al. (2016) provide direct evidence that identity can affect the screening technology itself. In selective cherry-picking markets such as labor markets, employers devote less attention to resumes with negatively stereotyped minority names, whereas in lemon-dropping markets such as rental housing, landlords scrutinize minority applicants more intensively. More broadly, background characteristics may determine which questions to ask, which benchmarks to use, and which additional signals are worth acquiring. In such cases, the relevant issue is not whether group identity or background should matter normatively, but how a decision-maker can elicit it truthfully when it is informative, unverifiable, and strategically manipulable.

I consider a manager who values group identity because it sharpens her prior about candidate quality and thereby changes the cost of learning whether the candidate is a good match. Candidates know their own group and can costlessly misreport it. The manager is rationally inattentive and acquires costly information about individual quality (Sims, 1998, 2003). Group identity therefore has purely instrumental value in the model: the manager ultimately cares only about match quality, but a more informative prior makes screening cheaper or more effective.

Without an ex ante commitment device, informative communication breaks down: in every equilibrium, the manager applies the same hiring rule after every report used on path. The problem is related to cheap talk (Crawford and Sobel, 1982), but the result is not immediate. For example, reporting a high-prior identity can have two opposing effects. It places the candidate in a more

¹For example, employers commonly use educational credentials to solicit and screen resumes (Rivera, 2011), and venture capitalists condition funding and valuation on founders' prior entrepreneurial experience and academic training (Hsu, 2007).

favorable pool, but it may also induce the manager to use a more demanding screening rule. A low-prior candidate might therefore be reluctant to mimic the high-prior report. What rules out such crossing incentives is monotonicity of the manager’s unconstrained screening rule: a higher interim belief weakly raises the probability of hiring in both match states. Hence, all candidates rank reports in the same direction, and any two reports used in equilibrium must induce the same state-contingent hiring rule.

The proposed solution is to allow for a report-contingent hiring quota, thus granting the manager commitment power. A quota is a publicly announced unconditional hiring probability for each reported category. It constrains how often the manager hires after each report, while still leaving her free to screen candidates within each reported category as she sees fit. This makes the commitment both limited and enforceable. The manager need not commit to a full state-contingent hiring rule, nor does anyone need to verify candidates’ true identities. Compliance can be checked at the aggregate level: a regulator or compliance office can verify ex post whether the promised fraction of applicants in each reported category was hired, even though the manager’s private signal and candidates’ true types are unobserved.

I derive three main results. First, under general posterior-separable information costs, I characterize when quotas can sustain truthful revelation. Truthful reporting is incentive compatible if and only if the quota difference lies in an interval determined endogenously by the manager’s screening behavior. Moreover, the quota-supported feasible set contains the pooling benchmark, so a manager who could select among quota policies would weakly prefer the best quota-supported outcome to the no-commitment outcome. A useful way to read these results is that quotas solve a problem that arises prior to the usual fairness or redistribution question. In this paper, the primary friction is not that the manager uses identity, but that she cannot credibly elicit identity when candidates can costlessly falsify it. Quotas matter because they create an enforceable form of partial commitment that can make truthful communication sustainable.

Second, I characterize when truthful revelation can be sustained without changing any candidate’s hiring probability relative to the pooling case. This question is important because a quota policy that improves the manager’s screening problem may still raise distributional concerns if it makes some candidates less likely to be hired than under the pooling benchmark. Replication is possible if and only if the pooled-prior hiring rule satisfies a replication condition: it must remain optimal under each group-specific prior when quotas are adjusted to match that rule. A global sufficient condition for this is a structural property of the cost function that I call Prior Invariance: any

hiring rule that is optimal under one prior-quota pair can be transported to any other prior by adjusting the quota alone. When Prior Invariance holds, the manager can elicit identity truthfully while keeping the same state-contingent hiring rule she would use under the pooled prior. When the replication condition fails at the pooled-prior rule, no quota policy can sustain truthful revelation while preserving the pooling hiring rule after both reports.

Third, I apply the analysis to two of the most commonly used cost functions in the rational inattention literature, entropy and quadratic, and show that entropy costs satisfy Prior Invariance whereas quadratic costs do not. Under entropy costs, quotas can restore truthful revelation as a pure cost-saving device: the manager implements the same hiring rule as under pooling, but with lower information costs because screening starts from sharper group-specific priors. Under quadratic costs, by contrast, replication generically fails. Quotas can still sustain truthful revelation and benefit the manager, but at least one report must induce a different hiring rule from pooling, so redistribution across types becomes part of the mechanism.

This perspective also clarifies the role of entropy in the paper. Entropy costs are canonical in the rational inattention literature because of their tractability and information-theoretic foundations (Sims, 2003; Matějka and McKay, 2015; Denti, 2023). Here, it also implies a strong invariance property: the manager can keep the same state-contingent hiring rule optimal across different priors by adjusting quotas appropriately. That feature is not generic within the broader class of posterior-separable costs. More broadly, recent work shows that prior dependence is often a natural feature of information costs once they are derived from more primitive sequential information-acquisition technologies (Bloedel and Zhong, 2022; Denti et al., 2022). For this reason, the entropy benchmark is best interpreted here as a useful special case rather than as a general prediction.

Related Literature. A natural benchmark is the literature on affirmative action and endogenous investment in environments with statistical discrimination. In these models, policy affects incentives to invest in skills, and because beliefs, assignment rules, and investment are jointly determined, its effects on inequality are ambiguous and can either mitigate or perpetuate discriminatory outcomes (Coate and Loury, 1993; Moro and Norman, 2003). More recent work studies equilibrium screening when candidates invest in skills and screeners choose how intensively to screen, sometimes also allowing candidates to change identity endogenously (Fosgerau et al., 2023; Kim and Loury, 2019).² Relatedly, Echenique and Li (2025) study statistical discrimination when decision-makers are

²Rozzi et al. (2023) also study group choice together with screening in the presence of additional forces such as homophily.

rationally inattentive and belief formation is endogenous. This paper abstracts from endogenous skill investment and endogenous identity choice to isolate a different friction: costless falsification of a fixed, unverifiable characteristic together with costly information acquisition by the screener. As a result, quotas here do not operate by changing ex ante investment incentives. Instead, they serve as an auditable commitment device that can sustain truthful revelation at the screening stage. In this respect, this paper is closer to [Matveenko and Mikhlishchev \(2021\)](#), who show that quotas affect a rationally inattentive manager’s information acquisition. This paper builds on that informational role of quotas by making reported identity strategic and unverifiable.

This paper is also related to cheap-talk models in which communication interacts with endogenous information acquisition. For example, [Pei \(2015\)](#), [Argenziano et al. \(2016\)](#), and [Kreutzkamp \(2023\)](#) study sender-side costly information acquisition, while [Ivanov and Sam \(2022\)](#) study a cheap-talk environment in which the signal structure is chosen by the receiver before communication takes place. The model shares the basic tension between information acquisition and credibility, but the timing and mechanism are different. Candidates know a fixed identity, can misreport it, and do not acquire information before communicating. Instead, costly information acquisition is on the receiver’s side: after observing a report, the manager optimally chooses how much to learn. Without commitment, this receiver-side reoptimization causes identity communication to unravel into pooling. Persuasion models with receiver-side inattention or costly learning likewise emphasize that the receiver’s information-acquisition problem shapes what can be achieved by persuasion ([Bloedel and Segal, 2020](#); [Matysková and Montes, 2023](#)). More broadly, the paper contributes to the literature on commitment in persuasion and information design ([Kamenica, 2019](#); [Bergemann and Morris, 2019](#)). However, in those environments the sender typically designs an information structure rather than sending a cheap-talk report. Here, quotas play the role of a limited commitment device. In this sense, the paper is also related to analyses of partial commitment in communication and persuasion, such as [Min \(2021\)](#) and [Lipnowski et al. \(2022\)](#). It is also related to [Bergemann et al. \(2023\)](#), who show that quota-like restrictions can be useful communication rules in persuasion problems. The setting differs in that agents send cheap-talk reports about a fixed identity, and the receiver uses quotas to make truthful revelation incentive compatible during screening.³

The paper is related to work on screening when agents can manipulate observed signals. [Perez-](#)

³Relatedly, some recent work studies why decision-makers may voluntarily refrain from using informative inputs. [Dessein et al. \(2025\)](#) study why colleges may choose not to observe test scores, while [Liang et al. \(2026\)](#) analyze when a designer may ban inputs because of disagreement with how a decision-maker would use them. Here, by contrast, the issue is not whether an informative attribute should be suppressed, but how it can be elicited truthfully.

Richet and Skreta (2022) analyze test design when workers can falsify outcomes, showing that optimal tests may deliberately coarsen or reshape distinctions to deter manipulation. Perez-Richet and Skreta (2023) extend this logic with fairness considerations. Frankel and Kartik (2022) show that a receiver who commits to using coarser information can improve outcomes by dampening incentives to game the system. Ball (2025) shows that a principal can reduce manipulation by distorting the weights placed on observable features in a scoring rule, rather than using each feature according to its direct predictive value. The mechanisms in these papers are similar in spirit to quotas, since in each case the principal voluntarily restricts how information is used to discipline strategic behavior. However, the manipulation target and the principal’s instrument are different. Candidates falsify a pre-existing characteristic rather than a test outcome or a score-relevant feature. Correspondingly, rather than redesigning the test or scoring rule, the manager commits to report-contingent acceptance rates that make truthful identity reporting incentive compatible.⁴

2 Model

There is a heterogeneous population of candidates, which is treated as a continuum. Each candidate (he) belongs to a group that corresponds to a particular probability of being a good match. Candidates can misreport their group identity. The HR manager (she) screens candidates one at a time and wants to hire a candidate only if he is a good fit for the job. She does not observe match quality $\omega \in \Omega = \{\omega_B, \omega_G\}$ ex ante, but can acquire costly information about it. Both the candidate’s and the manager’s problems are described in detail below.

2.1 The Candidates

Each candidate belongs to a group or category. The candidate’s type is encoded by $t \in \{L, H\}$. Conditional on type t , match quality is

$$\Pr(\omega = \omega_G | t) = g_t, \quad \Pr(\omega = \omega_B | t) = 1 - g_t,$$

⁴A related strand studies costly verification and auditing. In Erlanson and Kleiner (2020), the principal pays to verify information, and Carroll and Egorov (2019) show how randomized verification of a single dimension can extract information from multidimensional reports. The mechanism is different. The manager never verifies identity. Instead, she acquires information about match quality at a cost, and quotas operate as an aggregate commitment that disciplines reports without verification.

where $0 < g_L < g_H < 1$. Let $\sigma \in (0, 1)$ denote the population share of group L . A candidate receives a wage $0 < w < 1$ if hired and utility 0 otherwise. Therefore, each candidate strictly prefers a higher hiring probability. The wage w is assumed to be determined in a competitive market and cannot be negotiated. Each candidate's type is fixed and unobservable to the manager. However, the candidate can costlessly send a report $m \in \{L, H\}$ to the manager. Let $\mathbf{m} : \{L, H\} \rightarrow \{L, H\}$ denote the candidate's (pure) strategy that maps the true type t to the report $m(t)$.

Assumption 1. Candidates know their group identity t but do not observe their quality realization ω .

For example, job candidates know their background but may not be aware of how well-suited they are for a particular role; students know their demographics but may not be aware of their true academic ability.

2.2 The Manager

The manager faces a discrete choice $a \in A \equiv \{C, R\}$: whether to hire the candidate ($a = C$) or reject ($a = R$). Payoffs depend on the match quality $\omega \in \Omega = \{\omega_B, \omega_G\}$:

$$\begin{aligned} v(C|\omega_G) &= 1 - w, & v(C|\omega_B) &= -w, \\ v(R|\omega_G) &= 0, & v(R|\omega_B) &= 0. \end{aligned}$$

After observing a report $m \in \{L, H\}$ and before choosing a , the manager can choose any signal structure $\phi : \Omega \times \{L, H\} \rightarrow \Delta(S)$, which prescribes a distribution over signal realizations $s \in S$ for every pair $(\omega, m) \in \Omega \times \{L, H\}$, where the signal space S is finite but arbitrarily rich. Let $\bar{g} := \sigma g_L + (1 - \sigma)g_H$ denote the population prior on ω_G .⁵

Information acquisition is costly. Information costs are posterior separable (Caplin et al., 2022; Denti, 2022):

$$C(\phi, \mu(\cdot|m)) = \lambda \left[H(\mu(\omega_G|m)) - \sum_{s \in S} \left(\sum_{\omega \in \Omega} \mu(\omega|m) \phi(s|\omega, m) \right) H(\mu(\omega_G|m, s)) \right], \quad (1)$$

where $\lambda > 0$ is the unit cost of information and $H : [0, 1] \rightarrow \mathbb{R}$ is continuous and strictly concave

⁵The interim belief $\mu(\omega_G|m)$ incorporates the manager's equilibrium conjecture about the candidate's reporting strategy and report m . The posterior belief $\mu(\omega_G|m, s)$ further incorporates signal s from the manager's learning strategy. The notation μ is used throughout for scalar beliefs (prior, interim, or posterior), distinguished by their conditioning arguments.

on $[0, 1]$, and twice continuously differentiable on $(0, 1)$. The two standard special cases the paper considers are **entropy costs**, where for $x \in [0, 1]$, $H(x) = -x \log x - (1 - x) \log(1 - x)$, and **quadratic costs**, where $H(x) = x(1 - x)$.

It is well known in the rational inattention literature (see, e.g., [Matějka and McKay, 2015](#); [Caplin et al., 2022](#)) that the optimal strategy involves requesting a “recommendation” signal indicating the optimal action, then following it. Formally, the manager’s problem can be written as choosing, for each report m , a decision rule $P(\cdot|\cdot, m) : \Omega \rightarrow \Delta(A)$, rather than a signal structure together with a choice rule. Throughout the paper, the analysis therefore works directly with the state-contingent action probabilities $P(a|\omega, m)$, which denote the probability of action $a \in A$ in state $\omega \in \Omega$. Given a report m and interim belief $\mu(\cdot|m)$, the manager’s problem is

$$\max_{P(a|\omega, m)} \sum_{\omega \in \Omega} \sum_{a \in A} v(a|\omega) P(a|\omega, m) \mu(\omega|m) - C(P, \mu(\cdot|m)), \quad (2)$$

where $C(P, \mu(\cdot|m))$ denotes, with a slight abuse of notation, the information cost induced by the decision rule $P(\cdot|\cdot, m)$, defined as the minimum of $C(\phi, \mu(\cdot|m))$ over signal structures $\phi(s|\omega, m)$ that generate $P(\cdot|\cdot, m)$. Since $\Omega = \{\omega_B, \omega_G\}$ is binary, the prior distribution $\mu(\cdot|m)$ is equivalently summarized by the scalar $\mu(\omega_G|m)$.

2.3 Quota

A quota policy specifies a target unconditional hiring probability $q_m \in [0, 1]$ for each report $m \in \{L, H\}$, and the manager is committed to satisfying these targets ([Matveenko and Mikhilishchev, 2021](#)). Quotas apply to reported categories rather than true types. Since the population of candidates is modeled as a continuum, compliance is exact at the aggregate level within each reported category, and the manager’s feasible commitments are limited by assumption to these aggregate unconditional acceptance rates rather than to full report-contingent state rules. Formally, when the manager faces a quota q_m for report m and an interim belief $\mu(\omega|m)$, she chooses state-contingent probabilities $P(C|\omega, m)$ to maximize (2) subject to the constraint that the unconditional hiring probability given report m equals the quota:

$$\mu(\omega_G|m) P(C|\omega_G, m) + (1 - \mu(\omega_G|m)) P(C|\omega_B, m) = q_m. \quad (3)$$

The analysis treats the quota schedule as a commitment instrument and asks what equilibria and allocations can be supported by such a commitment. The analysis does not take a stand on whether the quota is chosen by the manager or a regulator. Different quota setters may select different points from the same IC-feasible set. For example, a manager may prefer a quota that maximizes her payoff, while a planner may select a quota with different distributive or welfare objectives. The results below characterize feasibility and implementation conditional on any quota schedule.

2.4 Timing

The game proceeds as follows:

1. A quota policy $\mathbf{q} = (q_L, q_H) \in [0, 1]^2$ is publicly announced, committing the manager to unconditional hiring probability q_m for candidates who report m .
2. For each candidate, nature draws type $t \in \{L, H\}$, with $\Pr(t = L) = \sigma$, and match quality ω , with $\Pr(\omega = \omega_G | t) = g_t$. The candidate observes his type t and the quota policy, but not his quality realization ω , then sends a report $m \in \{L, H\}$.
3. The manager observes the report m , updates her belief, and chooses state-contingent hiring probabilities $P(C|\omega, m)$ subject to the quota constraint (3).
4. Hiring decisions and payoffs are realized.

The manager chooses freely after observing reports in the “no commitment” benchmark.

2.5 Equilibrium

The solution concept is Perfect Bayesian equilibrium, defined as follows.

Definition 1 (Equilibrium). Fix a quota policy $\mathbf{q} = (q_L, q_H)$. An equilibrium of the game is characterized by $(\mathbf{m}^*, P^*, \mu)$: the candidate’s report strategy $\mathbf{m}^* : \{L, H\} \rightarrow \{L, H\}$, the manager’s state-contingent action probabilities $P^* = (P^*(a|\omega, m))_{a \in A, \omega \in \Omega, m \in \{L, H\}}$, and a belief system $\mu = (\mu(\omega_G|m))_{m \in \{L, H\}}$, such that:

1. For each report $m \in \{L, H\}$, the decision rule $P^*(\cdot|m)$ solves the manager’s problem (2) given $\mu(\omega_G|m)$ (and, in the quota regime, subject to the quota constraint associated with q_m);

2. For each type $t \in \{L, H\}$, $\mathbf{m}^*(t)$ maximizes the candidate's expected payoff,

$$w \left[g_t P^*(C|\omega_G, m) + (1 - g_t) P^*(C|\omega_B, m) \right], \quad (4)$$

over $m \in \{L, H\}$, given P^* ;

3. Beliefs are consistent with strategies: $\mu(\omega_G|m)$ satisfies Bayes' rule whenever possible given the primitives (σ, g_L, g_H) and the messaging strategy $\mathbf{m}^*$.

3 No Commitment

The benchmark is the case in which the manager cannot commit ex ante to any report-contingent hiring rule or information policy. After observing a report, she simply solves her unconstrained rational inattention problem given the interim belief induced by that report.

Because messaging is costless and candidates care only about being hired, one might expect all candidates to choose whichever report induces the most favorable continuation policy. The nontrivial point is that the ranking of continuation policies need not be obvious from the reported prior alone. A report associated with a high prior may lead the manager to hire more often on average, but it may also induce a more demanding screening rule. Thus, if the low-type candidate claims to be the high type, he may also expect to be evaluated more intensively, but, unlike a true high type, he is more likely to fail that scrutiny. A high type might then want to report low in order to face an easier test. Thus, the no-commitment result is not a direct cheap-talk pooling argument.

What rules out such crossing incentives is monotonicity of the unconstrained screening rule. In the binary rational-inattention problem, a higher interim belief weakly raises the probability of hiring in both states, $P(C|\omega_G)$ and $P(C|\omega_B)$. Intuitively, when the manager's interim belief becomes more favorable, hiring becomes more attractive relative to rejection. In the binary problem, this shifts the optimal recommendation rule toward the hiring action rather than changing candidates' rankings of reports in opposite directions. Thus, a higher-prior report may induce stricter screening in the sense that bad matches are filtered out more often relative to good matches, but it does not make either state less likely to receive a hire recommendation under the unconstrained optimum. Hence, the continuation rule after a higher-belief report is weakly more favorable for every candidate type, regardless of his true group. Once all types rank reports in the same direction, two different on-path reports cannot induce different continuation policies. Therefore, without commitment, every PBE is

a pooling equilibrium. Proposition 1 formalizes this result.

Definition 2 (Pooling equilibrium). A PBE is a pooling equilibrium if the manager’s state-contingent hiring rule is independent of the report on the equilibrium path:

$$P^*(C|\omega, m) = P^*(C|\omega, m')$$

for every $\omega \in \Omega$ and every pair of reports $m, m' \in m^*(\{L, H\})$.

Proposition 1. *In the absence of commitment, a pooling equilibrium exists. Moreover, every PBE is a pooling equilibrium.*

See [proof](#) on page 26.

3.1 The value of observing identity

Although informative communication is impossible without commitment, the loss of identity information is costly to the manager. If she could condition her screening policy on the candidate’s true group, she would screen under finer priors and would weakly improve upon the pooled-prior benchmark. This gap motivates the search for a commitment device that can elicit truthful identity reports. Proposition 2 formalizes the observation.

Let $V^U(\mu)$ denote the manager’s unconstrained value function under a generic prior μ : the maximum of (2) over all feasible decision rules $P(\cdot|\cdot)$.

Proposition 2.

$$\sigma V^U(g_L) + (1 - \sigma)V^U(g_H) \geq V^U(\sigma g_L + (1 - \sigma)g_H), \quad (5)$$

with strict inequality whenever V^U is strictly convex on the interval with endpoints g_L and g_H .

See [proof](#) on page 27.

4 Quotas as Commitment

This section characterizes the incentive constraints for truthful reporting, gives conditions under which interior IC-feasible quotas exist, and establishes general welfare properties of truthful revelation. The general results set up a dichotomy: whether truthful revelation can be achieved without changing

any candidate's hiring probability depends on whether the cost function satisfies a structural property called Prior Invariance.

First, define a separating equilibrium.

Definition 3 (Separating equilibrium). A PBE is separating if candidates report truthfully, $\mathbf{m}^*(t) = t$ for $t \in \{L, H\}$, and the manager's on-path beliefs satisfy $\mu(\omega_G|m) = g_m$ for $m \in \{L, H\}$.

Fix a quota schedule (q_L, q_H) . In a separating equilibrium, after report m the manager believes the report, so $\mu(\omega_G|m) = g_m$. She then solves the quota-constrained rational inattention problem under this belief and quota q_m , yielding an optimal state-contingent hiring rule $(P(C|\omega_G, m, q_m), P(C|\omega_B, m, q_m))$ satisfying (3). A candidate from true group t who reports m is hired with unconditional probability

$$P_t(C|m) = g_t P(C|\omega_G, m, q_m) + (1 - g_t) P(C|\omega_B, m, q_m).$$

For any report m , the quota constraint implies that a feasible state-contingent hiring rule can be summarized by a single spread

$$\Delta_m := P(C|\omega_G, m, q_m) - P(C|\omega_B, m, q_m).$$

Since the manager's belief after report m is g_m , the quota constraint gives

$$P(C|\omega_G, m, q_m) = q_m + (1 - g_m)\Delta_m, \quad P(C|\omega_B, m, q_m) = q_m - g_m\Delta_m.$$

Thus, choosing a quota-constrained hiring rule is equivalent to choosing the spread Δ_m . Under the maintained strict concavity of H , the quota-constrained objective is strictly concave in Δ_m .⁶ Hence, the optimal spread is unique. Since the report affects the manager's problem only through the induced prior g_m and the assigned quota q_m , the optimal spread is denoted by $\Delta^*(g_m, q_m)$.

Substituting the optimal spread into the hiring probability gives

$$P_t(C|m) = q_m + (g_t - g_m)\Delta^*(g_m, q_m). \tag{6}$$

⁶Indeed, the gross payoff is affine in Δ_m . The action-induced posteriors are

$$\mu_C(\Delta_m) = g_m + \frac{g_m(1 - g_m)}{q_m}\Delta_m, \quad \mu_R(\Delta_m) = g_m - \frac{g_m(1 - g_m)}{1 - q_m}\Delta_m.$$

Thus, the induced information cost is $\lambda[H(g_m) - q_m H(\mu_C(\Delta_m)) - (1 - q_m) H(\mu_R(\Delta_m))]$, which is strictly convex in Δ_m because H is strictly concave. Therefore, gross payoff net of information cost is strictly concave.

Equation (6) shows how quotas discipline reports. Truthful reporting gives each type the quota assigned to his group, whereas a deviation exposes the candidate to a screening rule calibrated to the wrong prior. Thus, quotas affect incentives not only through average hiring rates, but also through the mismatch between true type and reported-type screening.

4.1 Incentive Compatibility under Quotas

Truthful reporting requires

$$\text{IC-L : } P_L(C|L) \geq P_L(C|H), \quad (7)$$

$$\text{IC-H : } P_H(C|H) \geq P_H(C|L). \quad (8)$$

Lemma 1 (Band condition). *Truthful reporting is incentive compatible if and only if*

$$(g_H - g_L)\Delta^*(g_L, q_L) \leq q_H - q_L \leq (g_H - g_L)\Delta^*(g_H, q_H). \quad (9)$$

See [proof](#) on page 27.

The band condition requires the quota gap $q_H - q_L$ to be large enough that type H does not gain by reporting L , yet small enough that type L does not gain by reporting H . The width of this band is governed by the spread Δ . A larger spread means that a deviating candidate, whose true prior g_t differs from the reported prior g_m , faces a larger discrepancy between the quota q_m and his actual hiring probability. This relaxes IC- L by making it less attractive for type L to report H , but tightens IC- H by making it more attractive for type H to report L . Thus, more intensive screening changes both bounds of the IC-feasible band and does not unambiguously make truthful reporting easier.

Remark 1 (Corner quotas). If $q_m \in \{0, 1\}$, the quota constraint forces $P(C|\omega_G, m) = P(C|\omega_B, m)$, so $\Delta^*(g_m, q_m) = 0$. Thus, corner quotas cannot generate the screening wedge that disciplines reports. By Lemma 1, unequal corner quotas cannot sustain truthful reporting, while equal corner quotas sustain only a trivial equilibrium in which all candidates are always rejected or always hired. For this reason, the analysis below focuses on interior quotas, meaning $q_m \in (0, 1)$ for each report m .

For later use, it is helpful to distinguish hiring rules that actually use information from those

that do not. A state-contingent hiring rule P is called informative if

$$\Delta(P) := P(C|\omega_G) - P(C|\omega_B) > 0.$$

A hiring rule is interior if

$$0 < P(C|\omega_B) < P(C|\omega_G) < 1.$$

4.2 Existence of Separating Quotas

Lemma 1 characterizes incentive compatibility for a fixed quota schedule. The question is whether there exist interior quotas that satisfy both incentive constraints and therefore support truthful reporting.

Given the uniqueness observation above, the optimal spread is well defined for each interior prior-quota pair. Define the two IC boundary curves in $[0, 1]^2$:

$$\mathcal{C}_H := \{(q_L, q_H) : q_H = q_L + (g_H - g_L)\Delta^*(g_L, q_L)\}, \quad (10)$$

$$\mathcal{C}_L := \{(q_L, q_H) : q_H - q_L = (g_H - g_L)\Delta^*(g_H, q_H)\}. \quad (11)$$

The IC-feasible region consists of quota pairs satisfying the lower bound associated with $\mathcal{C}_H$ and the upper bound associated with $\mathcal{C}_L$.

Lemma 2 (Strict screening). *For any posterior-separable cost function, the manager acquires information:*

$$\Delta^*(\mu, q) > 0$$

for all $(\mu, q) \in (0, 1)^2$.

See [proof](#) on page 27.

At the boundary quotas, feasibility forces the spread to be zero: when $q = 0$, the constraint forces both conditional hiring probabilities to be 0, and when $q = 1$, it forces both to be 1. Together with Lemma 2, this implies that both IC boundary sets contain the corner points $(0, 0)$ and $(1, 1)$ and lie strictly above the diagonal $\{q_H = q_L\}$ for interior quotas.

The next proposition records the implementation step: any interior quota pair inside the IC-feasible region supports a separating equilibrium.

Proposition 3. *Suppose that there exist interior quotas $(q_L^*, q_H^*) \in (0, 1)^2$ that satisfy the band condition (9). Then there exists a separating equilibrium in which candidates report truthfully and, after report m , the manager applies an optimal quota-constrained hiring rule under (g_m, q_m^*) .*

See [proof](#) on page 27.

It remains to give a simple way to verify that such quotas exist. A convenient sufficient condition is obtained by starting from the lower IC boundary. Fix $q_L \in (0, 1)$ and set

$$q_H = q_L + (g_H - g_L)\Delta^*(g_L, q_L).$$

This makes type H 's incentive constraint bind. The remaining question is whether type L 's incentive constraint also holds at this quota pair. By Lemma 1, this is equivalent to

$$\Delta^*(g_H, q_H) \geq \Delta^*(g_L, q_L) \quad \text{where} \quad q_H = q_L + (g_H - g_L)\Delta^*(g_L, q_L). \quad (12)$$

Condition (12) is not automatic under posterior separability alone. It compares the screening spread at the initial pair (g_L, q_L) with the spread at the induced pair (g_H, q_H) . Whether this ordering holds depends on the specific cost function H . Under entropy costs, Prior Invariance (Section 5.1) ensures that this condition holds globally. Under quadratic costs (Section 5.2), it holds only for specific parameter values.

Corollary 1 (Sufficient condition for existence). *Suppose there exists $q_L \in (0, 1)$ such that, with*

$$q_H = q_L + (g_H - g_L)\Delta^*(g_L, q_L),$$

the induced quota satisfies $q_H \in (0, 1)$ and

$$\Delta^*(g_H, q_H) \geq \Delta^*(g_L, q_L).$$

Then (q_L, q_H) satisfies the band condition (9) and hence admits a separating equilibrium by Proposition 3.

See [proof](#) on page 28.

4.3 Replication of the Pooling Benchmark

Quotas cannot make the manager worse off when she is free to choose whether and how to use them. By setting equal quotas, she can reproduce the pooling equilibrium and attain the no-commitment pooling payoff.

Lemma 3. *There exists a quota policy under which the manager can attain her pooling payoff.*

See [proof](#) on page 28.

Quotas can also make truthful revelation feasible, allowing the manager to improve upon the pooling benchmark. There are two sources of potential gains. First, quotas may allow the manager to move hiring probabilities toward the unconstrained observed-identity benchmark. Second, even holding the state-contingent hiring rule fixed, truthful revelation may reduce information costs by letting the manager screen from sharper group-specific priors.

The replication question isolates this second channel. Can the manager elicit group identity while implementing the same hiring rule as under pooling, so that every candidate type's hiring probability is unchanged? This depends on whether the pooled-prior optimum can be transported to the group-specific priors by adjusting quotas alone. A global sufficient condition for this transport property is Prior Invariance.

Let P^p denote the manager's optimal hiring rule under the pooled prior $\bar{g}$. Define the replication quotas

$$q_t^{rep} := g_t P^p(C|\omega_G) + (1 - g_t) P^p(C|\omega_B), \quad t \in \{L, H\}. \quad (13)$$

If P^p is informative, then $q_L^{rep} < q_H^{rep}$ and both replication quotas are interior.

Definition 4 (Prior Invariance). A posterior-separable cost function satisfies Prior Invariance if, for any $(\mu, q) \in (0, 1)^2$, the optimal state-contingent hiring rule $(P^*(C|\omega_G, \mu, q), P^*(C|\omega_B, \mu, q))$ can be implemented as the optimum under any other prior $\mu' \in (0, 1)$ by setting the quota to

$$q' = \mu' P^*(C|\omega_G, \mu, q) + (1 - \mu') P^*(C|\omega_B, \mu, q).$$

The definition has the following interpretation. Holding the quota fixed, a change in the prior would generally change the manager's optimal hiring rule. Prior Invariance requires that this effect can be fully offset by adjusting the quota to the unconditional hiring probability induced by the original rule at the new prior. If so, the same state-contingent hiring rule remains optimal under the new prior μ' .

Definition 5 (Replication condition at a rule). Fix an informative state-contingent hiring rule P . P satisfies the replication condition across the group priors if, for each $t \in \{L, H\}$, the rule P is optimal under prior g_t and quota

$$q_t(P) := g_t P(C|\omega_G) + (1 - g_t) P(C|\omega_B).$$

Prior Invariance is a global property of the cost function. By contrast, Definition 5 is rule-specific and isolates the exact condition needed for replication of a given hiring rule across the group priors. Since P^p is optimal under the pooled prior and its induced quota

$$q^p = \bar{g} P^p(C|\omega_G) + (1 - \bar{g}) P^p(C|\omega_B),$$

global Prior Invariance implies that P^p satisfies the replication condition across the group priors. When the replication condition holds at P^p , the replication quotas (13) make P^p optimal under each group-specific prior, so every candidate type faces the same hiring rule as under pooling. When the replication condition fails, no separating equilibrium can preserve the pooling hiring rule after both reports. Sections 5.1 and 5.2 show that entropy costs satisfy Prior Invariance, whereas quadratic costs fail the replication condition at informative interior pooled rules.

Proposition 4 (Replication of the pooling benchmark). *Consider any posterior-separable cost function and suppose the benchmark hiring rule P^p is informative. The replication quotas (13) sustain a separating equilibrium in which the manager's on-path hiring rule is P^p after every report if and only if P^p satisfies the replication condition across the group priors.*

In any such equilibrium, all candidates' hiring probabilities are unchanged. In particular, if the cost function satisfies Prior Invariance, then the replication quotas sustain such a separating equilibrium. If P^p does not satisfy the replication condition across the group priors, then no quota policy simultaneously sustains truthful reporting and replicates P^p after both reports.

See [proof](#) on page 28.

In the replication equilibrium of Proposition 4, each type is indifferent across reports. This is sufficient for equilibrium: any arbitrarily small cost of falsification or intrinsic preference for honesty would make truthful reporting strictly optimal (Abeler et al., 2019; Kartik, 2009; Perez-Richet and Skreta, 2022). The proposition also delivers the paper's main allocation-neutrality result. Replication of the pooling rule succeeds exactly when P^p satisfies the replication condition across the group

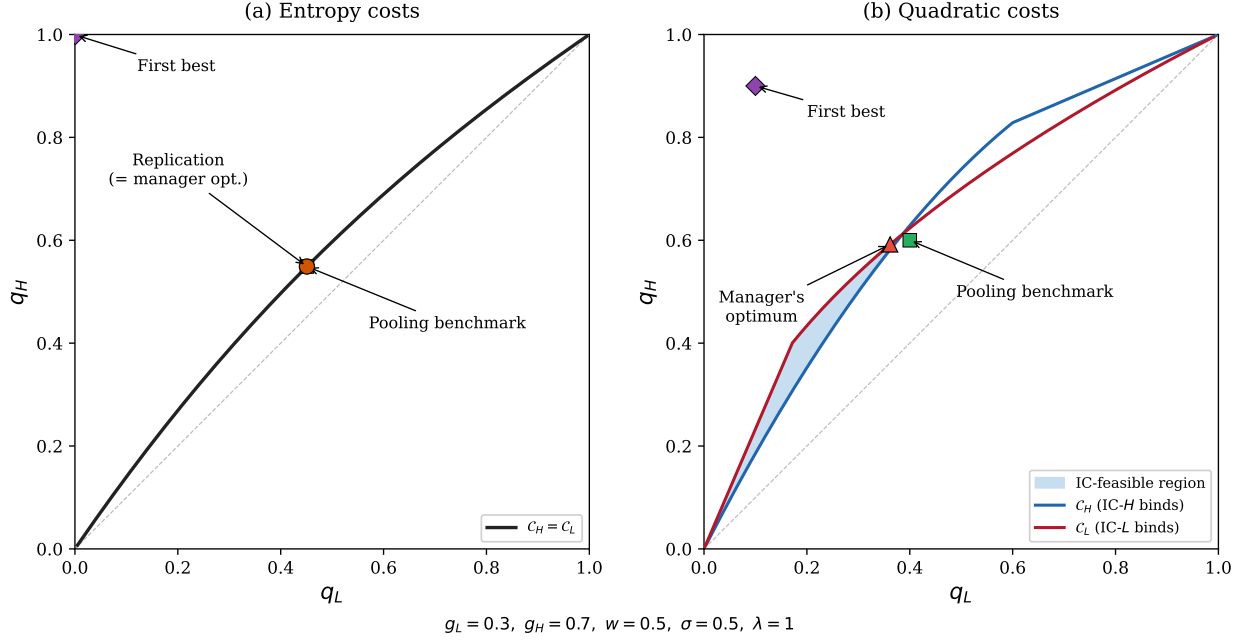


Figure 1: IC-feasible quotas under entropy and quadratic costs.

priors. Global Prior Invariance is a convenient sufficient condition for this rule-specific property. When the replication condition fails, no separating equilibrium can preserve the pooling hiring rule after both reports; any separating equilibrium must therefore use a hiring rule different from P^p after at least one report.

Figure 1 illustrates the IC-feasible set for the entropy (Section 5.1) and quadratic cost (Section 5.2) functions. In both panels, the dashed diagonal is the equal-quota line $q_H = q_L$, the square marker shows the pooling benchmark, evaluated separately for the two true groups, and the triangle shows the quota pair that maximizes the manager's ex ante payoff over the IC-feasible set for the displayed parameter values. The replication marker shows the quota pair q^{rep} defined in (13). The diamond marker shows the first best, i.e., the unconstrained optimum when the manager observes the group identity.

Under entropy costs, the group-specific priors $g_L = 0.3$ and $g_H = 0.7$ lie outside the screening region, so the first best is a corner solution: the manager rejects the group L candidates and hires the group H candidates based only on the prior. The two IC boundary curves coincide, so the IC-feasible set collapses to a one-dimensional curve along which both constraints bind. This is the Prior-Invariance case described in Proposition 4: once a quota adjustment transports an optimal rule from one prior to another, the same spread appears on both sides of the IC band. The

pooling benchmark, in the symmetric parameterization shown here, coincides with the manager’s payoff-maximizing quotas over the IC-feasible set.

Under quadratic costs, the first best is interior and lies outside the IC-feasible region. The two IC boundaries are distinct, so the feasible set is two-dimensional. The manager-preferred quota within the IC-feasible set lies on the upper boundary $\mathcal{C}_L$, so IC- L binds at the optimum. Intuitively, once quotas are sufficiently tilted to deter type H from mimicking L , the manager would like to increase the spread between q_H and q_L further, but is then constrained by the incentive of type L to mimic H .

5 Entropy and Quadratic Costs

5.1 Entropy Costs

This section specializes the analysis to entropy information costs (Shannon, 1948; Cover and Thomas, 2012). Under entropy costs, the manager’s optimal quota-constrained hiring rule satisfies a prior-independent odds-ratio condition, which implies Prior Invariance. Proposition 5 formalizes this result.

Proposition 5. *The entropy cost function satisfies Prior Invariance.*

See [proof](#) on page 29.

By Proposition 4, it follows that if the pooling hiring rule P^p is informative, then the replication quotas (13) sustain a separating equilibrium in which the manager’s on-path state-contingent hiring probabilities coincide with P^p after either report. Moreover, Lemma 4 shows that relative to the benchmark pooling equilibrium, while the candidates’ hiring probabilities are unchanged, the manager’s expected payoff strictly increases, yielding a Pareto improvement.

Lemma 4. *Under entropy costs, if P^p is informative, the replication quotas sustain a separating equilibrium that strictly increases the manager’s payoff relative to pooling.*

See [proof](#) on page 30.

The intuition for Prior Invariance under entropy is as follows. Shannon costs price information in “bits,” and the marginal cost of a bit does not depend on the prior. As a result, the optimal odds ratio between hiring in good and bad states is prior-invariant. By adjusting quotas across groups, the manager can therefore keep the same conditional hiring rule even though the priors differ. The

gain comes entirely from lower information costs: once identity is revealed truthfully, the manager starts from a sharper prior before screening, so the same interview standard or scoring rule has less uncertainty left to resolve and is cheaper to implement.⁷

A further implication is that the replication quotas make both IC constraints bind simultaneously. Each type is exactly indifferent across reports in the separating equilibrium.

The replication argument in Proposition 4 does not rely on Assumption 1. Because the manager uses the same state-contingent hiring rule after either report, a candidate who observed his quality realization ω would still be indifferent across reports in every state. Thus, under entropy costs, the allocation-neutral separating equilibrium supported by the replication quotas remains valid even if candidates observe quality before reporting. Moreover, the results extend directly to any finite number of groups (see Appendix B).

5.2 Quadratic Costs

Under quadratic costs, the manager’s optimal screening intensity depends on the prior even after conditioning on the quota. Proposition 6 shows that quadratic costs violate Prior Invariance. In particular, whenever the pooled-prior benchmark rule P^p is informative and interior, the replication condition fails at P^p across the group priors. Hence, by Proposition 4, no quota policy can sustain truthful reporting while replicating the pooling rule after both reports. Unlike under entropy costs, changing quotas alone cannot keep the same conditional hiring rule optimal across priors.

Proposition 6. *Under quadratic costs, Prior Invariance fails for any interior quota-constrained optimum. Moreover, if the pooled-prior hiring rule P^p is informative and interior, then P^p does not satisfy the replication condition across the group priors.*

See [proof](#) on page 31.

Proposition 7 then shows that interior separating quotas can exist in a nontrivial parameter region. As a result, quotas can still sustain truthful revelation and benefit the manager, but the gain need not be candidate-neutral and may come with redistribution across types. Because quadratic costs can generate corner solutions, the result imposes interiority conditions at the relevant quota-constrained optima.

⁷This Prior Invariance result is related to the invariant-likelihood-ratio property implied by Shannon rational inattention (Caplin and Dean, 2015) and to the locally invariant posteriors property (Caplin et al., 2022), which characterizes uniformly posterior-separable costs. The connection runs through Shannon specifically rather than through the uniformly posterior-separable class in general: in the present binary quota-constrained problem, Shannon costs imply a prior-independent odds ratio between hiring in the good and bad states. Because the quota fixes the unconditional hiring probability, that odds-ratio condition yields Prior Invariance.

Proposition 7. *Define*

$$\gamma_t := \frac{g_H - g_L}{2\lambda g_t(1 - g_t)}, \quad t \in \{L, H\}. \quad (14)$$

If

$$\gamma_H(1 + \gamma_L) > \gamma_L \quad \text{and} \quad \lambda > \frac{1}{2(1 - g_H)}, \quad (15)$$

then there exist interior quotas $(q_L^, q_H^*) \in (0, 1)^2$ satisfying both IC constraints at which the quota-constrained optima are interior, and hence admit a separating equilibrium. Moreover, any such pair satisfies $q_L^* < q_H^*$.*

See [proof](#) on page 34.

Under quadratic costs, the welfare consequences of separation are generally parameter-dependent. In contrast to the entropy case, restoring truthful revelation need not be candidate-neutral and typically requires a change in the allocation. Appendix C illustrates this point in a symmetric benchmark, where separating quotas strictly benefit the manager, but redistribute hiring probability across candidate types. Outside the symmetric benchmark, the welfare effects of separation cannot be signed in general. When $\sigma \neq 1/2$ or $w \neq \bar{g}$, pooling already differentiates types through the common screening rule, and the IC constraints can prevent separating quotas from matching that differentiation. In such cases, both types may be worse off under separation even when the manager benefits.

6 Conclusion

This paper studies how commitment over aggregate hiring rates can restore truthful revelation when candidates may strategically misreport a group characteristic that is informative for screening. Without commitment, truthful communication collapses: in equilibrium, the manager applies the same hiring rule after every equilibrium report, so reports carry no information and the allocation is effectively report-blind. Quotas change this by making continuation policies report-contingent. Under posterior-separable costs, truthful revelation can be sustained whenever the quota gap lies in the incentive-compatible band determined by the screening spreads at the group-specific priors. The paper then shows that whether this restoration of truthful communication can be achieved without changing the allocation depends sharply on the information cost. Under entropy costs, quotas can replicate the no-commitment allocation while inducing truthful revelation, so they operate as a pure revelation device, lowering costs without distorting the allocation. Under quadratic costs,

by contrast, restoring truthful revelation generally changes the allocation and redistributes hiring probabilities across types.

There is an alternative way to interpret the no-commitment benchmark. Because all equilibria are pooling, the manager ends up using the same hiring rule regardless of the reported group. In that sense, the no-commitment outcome resembles a blind-review or identity-blind policy: final hiring depends only on match quality, not on the reported characteristic (Goldin and Rouse, 2000). Quotas then act as a device that reintroduces group-contingent screening. This interpretation connects the paper to Matveenko and Mikhlishchev (2021), but also clarifies what is different here. In their model, group identity is observable, and quotas can be chosen to implement a fair allocation in which hiring probabilities depend only on job-relevant characteristics. In that sense, quotas can generate outcomes that are “as if” a blind or de-identified policy were in place, while still allowing the manager to use the information contained in group identity.

In this paper, group identity is not verifiable and must first be elicited. Quotas therefore play an additional role: they make report-contingent screening credible enough to sustain truthful revelation. Once truthful revelation is restored, the relation to the blind benchmark depends on the information-cost technology. Under entropy costs, the result is closest to Matveenko and Mikhlishchev (2021): quotas allow the manager to use truthful identity information without changing candidates’ hiring probabilities relative to the pooling benchmark. Under quadratic costs, this allocation-neutral implementation generally fails. Consequently, any quota policy that restores truthful revelation must also change the state-contingent hiring rule after at least one report, so the policy necessarily changes the allocation across types. Thus, the same policy instrument can have very different implications depending on the underlying attention technology: it may serve either as a purely informational commitment device or as a mechanism that necessarily redistributes hiring probabilities.

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A Proofs

Proof of Proposition 1. Existence. Let all candidates send the same report $\hat{m}$. Assign beliefs $\mu(\omega_G|m) = \bar{g}$ for every $m \in \{L, H\}$, and let the manager solve the unconstrained problem (2) under prior $\bar{g}$ after any report. Since the manager's hiring rule is then identical after L and H , every candidate is indifferent across reports, so sending $\hat{m}$ is optimal. The manager's screening behavior is optimal by construction, and beliefs are Bayes-consistent on path. Hence, this is a PBE, and it is pooling.

Necessity. Fix any PBE $(\mathbf{m}^*, P^*, \mu)$. For a generic prior $p \in [0, 1]$, the unconstrained binary-state, binary-action rational inattention problem is determined by the concavification of

$$f(p) := \max\{p - w, 0\} + \lambda H(p)$$

(see, e.g., Matějka and McKay, 2015; Caplin et al., 2022). Since H is strictly concave, the only possible nonconcavity of f comes from the kink at $p = w$. Therefore, the concave envelope of f coincides with f except possibly on a single interval $[\underline{p}^*, \bar{p}^*]$ with $\underline{p}^* \leq w \leq \bar{p}^*$. For priors $p \leq \underline{p}^*$ the manager always rejects; for $p \geq \bar{p}^*$ she always hires; for $p \in (\underline{p}^*, \bar{p}^*)$ she randomizes between posteriors $\underline{p}^*$ (reject) and $\bar{p}^*$ (hire). In this interior region, Bayes' plausibility gives

$$P^*(C|\omega_G, p) = \frac{\bar{p}^*(p - \underline{p}^*)}{p(\bar{p}^* - \underline{p}^*)} \quad \text{and} \quad P^*(C|\omega_B, p) = \frac{(1 - \bar{p}^*)(p - \underline{p}^*)}{(1 - p)(\bar{p}^* - \underline{p}^*)}, \quad (16)$$

which are weakly increasing in p on this interval. Combining this with the degenerate regions, for any two priors $p \leq p'$,

$$P^*(C|\omega_G, p') \geq P^*(C|\omega_G, p) \quad \text{and} \quad P^*(C|\omega_B, p') \geq P^*(C|\omega_B, p). \quad (17)$$

Now, take any two reports $m, m' \in m^*(\{L, H\})$ used on the equilibrium path, and suppose, without loss of generality, that $\mu(\omega_G|m) \geq \mu(\omega_G|m')$. By (17),

$$P^*(C|\omega_G, m) \geq P^*(C|\omega_G, m') \quad \text{and} \quad P^*(C|\omega_B, m) \geq P^*(C|\omega_B, m').$$

Because m' is used on path, there exists some type $t \in \{L, H\}$ such that $m^*(t) = m'$. Since both messages are available to every type, that type's hiring probability after m , given by (4), is weakly larger than his hiring probability after m' . Since reporting is costless and m' is used in equilibrium,

that type must be unwilling to deviate from m' to m . Hence,

$$g_t P^*(C|\omega_G, m) + (1 - g_t) P^*(C|\omega_B, m) = g_t P^*(C|\omega_G, m') + (1 - g_t) P^*(C|\omega_B, m').$$

As $0 < g_t < 1$, and each term on the left is weakly larger than its counterpart on the right, this equality is possible only if

$$P^*(C|\omega_G, m) = P^*(C|\omega_G, m') \quad \text{and} \quad P^*(C|\omega_B, m) = P^*(C|\omega_B, m').$$

Thus, any two on-path reports induce the same state-contingent hiring rule. Therefore, every PBE is a pooling equilibrium in the sense of Definition 2. $\square$

Proof of Proposition 2. By Caplin et al. (2022), under posterior-separable costs the manager's unconstrained value function $V^U(\mu)$ is convex in the prior μ . Therefore,

$$\sigma V^U(g_L) + (1 - \sigma) V^U(g_H) \geq V^U(\sigma g_L + (1 - \sigma) g_H)$$

by Jensen's inequality. The strict inequality holds whenever V^U is strictly convex in the interval with endpoints g_L and g_H . $\square$

Proof of Lemma 1. From (6)

$$P_L(C|L) = q_L, \quad P_H(C|H) = q_H,$$

$$P_L(C|H) = q_H - (g_H - g_L) \Delta^*(g_H, q_H), \quad P_H(C|L) = q_L + (g_H - g_L) \Delta^*(g_L, q_L).$$

IC- L is equivalent to the upper bound in (9), and IC- H is equivalent to the lower bound. $\square$

Proof of Lemma 2. See Proposition 7 in Matveenko and Mikhlishchev (2021). The conditions of Proposition 7 are satisfied here because payoff differences are state-dependent and posterior-separable costs have zero marginal cost at uninformative policies. $\square$

Proof of Proposition 3. Fix quotas (q_L^*, q_H^*) and consider truthful reporting and on-path beliefs that assign probability one to the reported group. Given any report m , the manager's optimal hiring rule solves her quota-constrained problem under (g_m, q_m^*) , so her behavior is sequentially rational. Since (q_L^*, q_H^*) satisfies the band condition (9), truthful reporting is optimal for both candidate types by

Lemma 1. Beliefs are Bayes-consistent on path. Hence, this is a separating equilibrium in the sense of Definition 3. $\square$

Proof of Corollary 1. By construction,

$$q_H - q_L = (g_H - g_L)\Delta^*(g_L, q_L),$$

so the lower bound in Lemma 1 holds with equality. The condition

$$\Delta^*(g_H, q_H) \geq \Delta^*(g_L, q_L)$$

implies

$$q_H - q_L \leq (g_H - g_L)\Delta^*(g_H, q_H),$$

so the upper bound in Lemma 1 also holds. Hence, (q_L, q_H) satisfies the band condition (9). The existence of a separating equilibrium then follows from Proposition 3. $\square$

Proof of Lemma 3. Let P^p denote the manager's unconstrained optimal hiring rule under the pooled prior $\bar{g}$, and let

$$q^p := \bar{g}P^p(C|\omega_G) + (1 - \bar{g})P^p(C|\omega_B)$$

be the unconditional hiring probability it induces. Consider the quota policy $q_L = q_H = q^p$. Under this policy, a pooling equilibrium exists in which all candidates send the same report, and the manager applies P^p after any report. Since P^p satisfies the quota constraint with quota q^p and is optimal in the larger unconstrained problem under prior $\bar{g}$, it is also optimal in the constrained problem with quota q^p . Hence, the manager can attain her pooling payoff under quotas. $\square$

Proof of Proposition 4. Because P^p is informative, the replication quotas are interior and satisfy $q_L^{rep} < q_H^{rep}$.

First suppose that P^p satisfies the replication condition across the group priors. Then, for each $t \in \{L, H\}$, the rule P^p is optimal under prior g_t and quota q_t^{rep} . Consider truthful reporting and on-path beliefs that assign probability one to the reported group. After report t , the manager applies P^p , which is sequentially optimal under (g_t, q_t^{rep}) .

Since the manager applies the same state-contingent rule P^p after both reports, a type- t

candidate's hiring probability from any report is

$$g_t P^p(C|\omega_G) + (1 - g_t) P^p(C|\omega_B).$$

Thus, every type is indifferent across reports, so truthful reporting is a best response. Beliefs are Bayes-consistent on path. Therefore, the replication quotas sustain a separating equilibrium in which the on-path hiring rule is P^p after every report. In this equilibrium, each type's hiring probability is exactly the same as under pooling.

Conversely, suppose that the replication quotas sustain a separating equilibrium in which the manager's on-path hiring rule is P^p after both reports. In a separating equilibrium, after report t the manager's belief is g_t . Since the manager applies P^p after report t , sequential rationality requires P^p to be optimal under prior g_t and quota q_t^{rep} . This must hold for both $t = L$ and $t = H$. Hence, P^p satisfies the replication condition across the group priors.

If the cost function satisfies Prior Invariance, then P^p satisfies the replication condition because P^p is optimal under the pooled prior and its induced quota

$$q^p = \bar{g} P^p(C|\omega_G) + (1 - \bar{g}) P^p(C|\omega_B),$$

and Prior Invariance transports this rule to each group prior by setting the quota equal to q_t^{rep} .

Finally, suppose some quota policy sustains truthful reporting and replicates P^p after both reports. Fix $t \in \{L, H\}$. If the manager plays P^p after report t and believes the report on path, then the quota constraint after report t must hold under prior g_t :

$$g_t P^p(C|\omega_G) + (1 - g_t) P^p(C|\omega_B) = q_t.$$

Thus, necessarily $q_t = q_t^{rep}$. Therefore, the replication quotas are the only quotas that can replicate P^p after both reports. If P^p does not satisfy the replication condition across the group priors, then P^p is not optimal under at least one of (g_t, q_t^{rep}) , contradicting sequential rationality. Hence, no quota policy can simultaneously sustain truthful reporting and replicate P^p after both reports. $\square$

Proof of Proposition 5. Fix $(\mu, q) \in (0, 1)^2$. By [Matveenko and Mikhilishchev \(2021, Proposition 1\)](#), under entropy costs with quotas $(q, 1 - q)$ over actions $\{C, R\}$, the optimal conditional choice

probabilities satisfy

$$P(a|\omega) = \frac{q_a \exp((v(a|\omega) - \varphi_a)/\lambda)}{\sum_{a' \in \{C, R\}} q_{a'} \exp((v(a'|\omega) - \varphi_{a'})/\lambda)},$$

where $q_C := q$ and $q_R := 1 - q$ are the unconditional action probabilities, and φ_a is the Lagrange multiplier on the quota constraint for action a . Specializing to $a = C$ and using $v(R|\omega) = 0$, it follows that

$$P(C|\omega, \mu, q) = \frac{q \exp((v(C|\omega) - \varphi_C)/\lambda)}{q \exp((v(C|\omega) - \varphi_C)/\lambda) + (1 - q) \exp(-\varphi_R/\lambda)}.$$

Hence, the odds ratio across states satisfies

$$\frac{P(C|\omega_G, \mu, q)/(1 - P(C|\omega_G, \mu, q))}{P(C|\omega_B, \mu, q)/(1 - P(C|\omega_B, \mu, q))} = \exp\left(\frac{v(C|\omega_G) - v(C|\omega_B)}{\lambda}\right) = e^{1/\lambda},$$

which is independent of (μ, q) .

Now fix any other prior $\mu' \in (0, 1)$, and define

$$q' = \mu' P^*(C|\omega_G, \mu, q) + (1 - \mu') P^*(C|\omega_B, \mu, q).$$

By construction, the pair $(P^*(C|\omega_G, \mu, q), P^*(C|\omega_B, \mu, q))$ satisfies the quota constraint under (μ', q') . Moreover, it satisfies the same prior-independent odds-ratio condition derived above. Under entropy costs, the quota-constrained optimum is uniquely characterized by the quota constraint together with this odds-ratio condition (Matveenko and Mikhlishchev, 2021). Therefore, $(P^*(C|\omega_G, \mu, q), P^*(C|\omega_B, \mu, q))$ is also optimal under (μ', q') . This proves Prior Invariance. $\square$

Proof of Lemma 4. By Proposition 5, entropy costs satisfy Prior Invariance. Hence, by Proposition 4, the replication quotas sustain a separating equilibrium in which the manager applies P^p after both reports. Therefore, the state-contingent hiring rule, and hence the manager's gross expected payoff, is the same as under pooling.

It remains to compare information costs. Mutual information is concave in the prior distribution for any fixed channel (Cover and Thomas, 2012). Since P^p is informative, this concavity is strict, so

$$\sigma C(P^p, g_L) + (1 - \sigma) C(P^p, g_H) < C(P^p, \bar{g}).$$

Thus, expected information costs are strictly lower under truthful separation than under pooling, while gross payoffs are unchanged. Therefore, the manager's payoff strictly increases relative to pooling. $\square$

Proof of Proposition 6. Under quadratic costs, $H(\mu) = \mu(1 - \mu)$, so the posterior-separable cost of information is

$$C(P, \mu) = \lambda [\mu(1 - \mu) - \mathbb{E}[\mu^{post}(1 - \mu^{post})]] = \lambda \text{Var}(\mu^{post}), \quad (18)$$

where μ^{post} denotes the posterior belief induced by the manager's information policy. The second equality follows because

$$\mu(1 - \mu) - \mathbb{E}[\mu^{post}(1 - \mu^{post})] = \mathbb{E}[(\mu^{post})^2] - (\mathbb{E}[\mu^{post}])^2 = \text{Var}(\mu^{post}),$$

using $\mathbb{E}[\mu^{post}] = \mu$.

Fix $(\mu, q) \in (0, 1)^2$ and consider the quota constraint

$$\mu P(C|\omega_G) + (1 - \mu)P(C|\omega_B) = q.$$

Parameterize the decision rule by the spread Δ as defined in Section 4:

$$\Delta := P(C|\omega_G) - P(C|\omega_B) \geq 0.$$

The quota constraint implies

$$P(C|\omega_G) = q + (1 - \mu)\Delta, \quad P(C|\omega_B) = q - \mu\Delta. \quad (19)$$

Feasibility requires $P(C|\omega_G), P(C|\omega_B) \in [0, 1]$, which is equivalent to

$$0 \leq \Delta \leq \Delta^{max}(\mu, q) := \min \left\{ \frac{q}{\mu}, \frac{1 - q}{1 - \mu} \right\}. \quad (20)$$

With $|A| = |\Omega| = 2$, any feasible pair can be implemented by a binary recommendation signal $s \in \{C, R\}$ that the manager follows. Let μ^C and μ^R denote the posteriors after recommendations C and R . Bayes' rule gives

$$\mu^C = \frac{\mu P(C|\omega_G)}{q}, \quad \mu^R = \frac{\mu(1 - P(C|\omega_G))}{1 - q}.$$

Using (19),

$$\mu^C - \mu = \frac{\mu(1 - \mu)\Delta}{q}, \quad \mu^R - \mu = -\frac{\mu(1 - \mu)\Delta}{1 - q}.$$

Therefore,

$$C(\Delta, \mu, q) = \lambda \text{Var}(\mu^{post}) = \lambda(q(\mu^C - \mu)^2 + (1-q)(\mu^R - \mu)^2) = \lambda \frac{[\mu(1-\mu)]^2 \Delta^2}{q(1-q)}. \quad (21)$$

The manager's gross expected payoff from hiring equals

$$\mu(1-w)P(C|\omega_G) - w(1-\mu)P(C|\omega_B) = q(\mu - w) + \mu(1-\mu)\Delta,$$

so her quota-constrained problem reduces to choosing $\Delta \in [0, \Delta^{max}(\mu, q)]$ to maximize

$$V(\Delta, \mu, q) = q(\mu - w) + \mu(1-\mu)\Delta - \lambda \frac{[\mu(1-\mu)]^2 \Delta^2}{q(1-q)}. \quad (22)$$

The objective is strictly concave in Δ . The first-order condition yields

$$\Delta^{FOC}(\mu, q) = \frac{q(1-q)}{2\lambda\mu(1-\mu)}. \quad (23)$$

For an interior optimum, it follows that $\Delta^*(\mu, q) = \Delta^{FOC}(\mu, q) \in (0, \Delta^{max}(\mu, q))$.

It remains to verify the failure of Prior Invariance. Fix (μ, q) with an interior optimum, and let

$$\Delta^*(\mu, q) := P^*(C|\omega_G, \mu, q) - P^*(C|\omega_B, \mu, q) \in (0, 1).$$

For any other prior $\mu' \in (0, 1)$, Prior Invariance would require that the same pair

$$(P^*(C|\omega_G, \mu, q), P^*(C|\omega_B, \mu, q))$$

is optimal under the transported quota

$$q' = \mu' P^*(C|\omega_G, \mu, q) + (1 - \mu') P^*(C|\omega_B, \mu, q).$$

Using (19) and the identities

$$P^*(C|\omega_B, \mu, q) = q - \mu\Delta^*(\mu, q), \quad P^*(C|\omega_G, \mu, q) = q + (1 - \mu)\Delta^*(\mu, q),$$

it follows that

$$q' = q + (\mu' - \mu)\Delta^*(\mu, q).$$

Under quadratic costs with an interior optimum at (μ', q') , optimality requires

$$\Delta^*(\mu, q) = \Delta^{FOC}(\mu', q') = \frac{q'(1 - q')}{2\lambda\mu'(1 - \mu')}.$$

Equivalently, for all $\mu' \in (0, 1)$,

$$q'(1 - q') = 2\lambda\Delta^*(\mu, q)\mu'(1 - \mu'). \quad (24)$$

Write $q' = a + \Delta^*(\mu, q)\mu'$ where

$$a := q - \mu\Delta^*(\mu, q) = P^*(C|\omega_B, \mu, q).$$

Then

$$\begin{aligned} q'(1 - q') &= (a + \Delta^*(\mu, q)\mu')(1 - a - \Delta^*(\mu, q)\mu') \\ &= a(1 - a) + \Delta^*(\mu, q)(1 - 2a)\mu' - (\Delta^*(\mu, q))^2(\mu')^2. \end{aligned}$$

The right side of (24) equals

$$2\lambda\Delta^*(\mu, q)\mu' - 2\lambda\Delta^*(\mu, q)(\mu')^2.$$

Matching coefficients in μ' yields three conditions:

$$\text{constant: } a(1-a) = 0, \quad \text{linear: } \Delta^*(\mu, q)(1-2a) = 2\lambda\Delta^*(\mu, q), \quad \text{quadratic: } (\Delta^*(\mu, q))^2 = 2\lambda\Delta^*(\mu, q).$$

Since $\Delta^*(\mu, q) > 0$, the quadratic condition implies $\Delta^*(\mu, q) = 2\lambda$. The constant condition implies $a \in \{0, 1\}$. Since $a = P^*(C|\omega_B, \mu, q)$, this requires $P^*(C|\omega_B, \mu, q) \in \{0, 1\}$, which contradicts the assumption that the optimum is interior. Therefore, (24) cannot hold for all μ' , so Prior Invariance fails.

It remains to verify the replication-condition statement for the pooled-prior rule. Let P^p be interior, and write

$$a := P^p(C|\omega_B), \quad \Delta := P^p(C|\omega_G) - P^p(C|\omega_B) > 0.$$

For any prior ν , the transported quota associated with this fixed rule is

$$q(\nu) = a + \Delta\nu.$$

Under quadratic costs, if the same rule P^p is an interior optimum under prior ν and quota $q(\nu)$, the first-order condition requires

$$\Delta = \frac{q(\nu)(1 - q(\nu))}{2\lambda\nu(1 - \nu)}.$$

Equivalently,

$$a(1 - a) + \Delta(1 - 2a - 2\lambda)\nu + \Delta(2\lambda - \Delta)\nu^2 = 0.$$

Because P^p is an interior optimum at the pooled prior $\bar{g}$, this equation holds at $\nu = \bar{g}$. If P^p satisfied the replication condition across the group priors, then the same equation would also hold at $\nu = g_L$ and $\nu = g_H$. Since $g_L < \bar{g} < g_H$, the quadratic polynomial above would have three distinct roots. Hence, it would have to be identically zero. In particular, its constant term would satisfy $a(1 - a) = 0$, contradicting interiority. Therefore, P^p does not satisfy the replication condition across the group priors under quadratic costs. $\square$

Proof of Proposition 7. When the quota-constrained optimal hiring rules are interior, Lemma 1 together with (23) implies that incentive compatibility is equivalent to

$$q_L \geq q_H - \gamma_H q_H(1 - q_H), \tag{25}$$

$$q_H \geq q_L + \gamma_L q_L(1 - q_L). \tag{26}$$

Fix $q_L \in (0, 1)$ and set

$$q_H := q_L + \gamma_L q_L(1 - q_L). \tag{27}$$

Then (26) holds with equality, and $q_H > q_L$.

It remains to verify (25), which is equivalent to

$$\gamma_H q_H(1 - q_H) \geq \gamma_L q_L(1 - q_L). \tag{28}$$

Let $q_L \downarrow 0$. From (27),

$$q_H = q_L(1 + \gamma_L) - \gamma_L q_L^2, \quad q_H(1 - q_H) = q_L(1 + \gamma_L) + O(q_L^2).$$

Substituting into (28) yields

$$\gamma_H q_H(1 - q_H) - \gamma_L q_L(1 - q_L) = q_L(\gamma_H(1 + \gamma_L) - \gamma_L) + O(q_L^2).$$

Under (15), the coefficient on q_L is strictly positive. Hence, there exists $\bar{q} \in (0, 1)$ such that (28) holds for all $q_L \in (0, \bar{q})$.

It remains to verify that the optimal hiring rules are interior. For any (μ, q) with q near 0, $\Delta^{max}(\mu, q) = q/\mu$, so

$$\frac{\Delta^{FOC}(\mu, q)}{\Delta^{max}(\mu, q)} = \frac{1 - q}{2\lambda(1 - \mu)} \xrightarrow{q \downarrow 0} \frac{1}{2\lambda(1 - \mu)}.$$

Since $\lambda > 1/(2(1 - g_H))$ and $1 - g_L > 1 - g_H$, both limits are strictly less than 1 for $\mu \in \{g_L, g_H\}$, so $\Delta^{FOC}(\mu, q) < \Delta^{max}(\mu, q)$ for all q sufficiently small. Similarly, $P(C|\omega_G, \mu, q) = q + (1 - \mu)\Delta^{FOC} \rightarrow 0$ as $q \downarrow 0$, so $P(C|\omega_G) < 1$ for small q . Moreover,

$$P(C|\omega_B, \mu, q) = q - \mu\Delta^{FOC}(\mu, q) = q \left[1 - \frac{1 - q}{2\lambda(1 - \mu)} \right].$$

Since $\lambda > \frac{1}{2(1 - g_H)}$ and $\mu \in \{g_L, g_H\}$, it follows that

$$2\lambda(1 - \mu) \geq 2\lambda(1 - g_H) > 1,$$

so $P(C|\omega_B, \mu, q) > 0$ for all sufficiently small q . Hence, there exists $\tilde{q} \in (0, \bar{q})$ such that the optimal hiring rules are interior at both (g_L, q_L) and (g_H, q_H) for all $q_L \in (0, \tilde{q})$.

Choose $q_L^* \in (0, \tilde{q})$ and define q_H^* by (27). Both IC constraints hold, and the optima are interior. Finally, any interior quota pair satisfying the quadratic IC constraints must satisfy

$$q_H \geq q_L + \gamma_L q_L(1 - q_L) > q_L,$$

so $q_L < q_H$. Since these quotas are interior and satisfy both IC constraints, Proposition 3 implies that they admit a separating equilibrium. $\square$

B Multigroup replication under entropy

Suppose that there is a finite set of groups $k \in K$ with priors $g_k \in (0, 1)$ and population shares $\sigma_k \in (0, 1)$ with $\sum_{k \in K} \sigma_k = 1$. Let the pooled prior be

$$\bar{g} = \sum_{k \in K} \sigma_k g_k,$$

and let P^P be the manager's benchmark hiring rule under the pooled prior.

Define, for each group $k \in K$, the replication quota

$$q_k^{rep} = g_k P^p(C|\omega_G) + (1 - g_k) P^p(C|\omega_B).$$

Proposition 8. *Under entropy costs, the quota policy $(q_k^{rep})_{k \in K}$ admits a separating equilibrium in which each candidate reports his true group and, after every report, the manager's on-path state-contingent hiring probabilities coincide with P^p .*

Proof. Fix $P^p(C|\omega_G)$ and $P^p(C|\omega_B)$. For each group k , the quota q_k^{rep} makes $(P^p(C|\omega_G), P^p(C|\omega_B))$ feasible under prior g_k . By Proposition 5, this same pair is optimal under (g_k, q_k^{rep}) . Therefore, after every report, the manager's hiring rule is P^p . A type- k candidate's hiring probability is then

$$g_k P^p(C|\omega_G) + (1 - g_k) P^p(C|\omega_B)$$

regardless of the report. Thus, all types are indifferent across reports, and reporting truthfully is optimal. Beliefs are Bayes-consistent on path. $\square$

C Symmetric welfare benchmark under quadratic costs

Proposition 9 (Welfare under quadratic costs: symmetric case). *Assume $g_H = 1 - g_L$, $\sigma = 1/2$, $w = 1/2$, and $\lambda > 1/2$. Consider any separating equilibrium in which both IC constraints bind, and suppose the quota-constrained optima at the separating quotas are interior.*

- (i) *Type L is worse off and type H is better off than under pooling.*
- (ii) *Average candidate welfare is unchanged.*
- (iii) *The manager's expected payoff is strictly higher than under pooling.*

Proof. Pooling benchmark. Under symmetry, the pooled prior is $\bar{g} = 1/2$. With $\lambda > 1/2$, the pooling optimum is interior and satisfies

$$\Delta^{FOC}(1/2, 1/2) = \frac{1}{2\lambda}.$$

Substituting $\mu = q = 1/2$ into (19) gives

$$P^p(C|\omega_G) = \frac{1}{2} + \frac{1}{4\lambda}, \quad P^p(C|\omega_B) = \frac{1}{2} - \frac{1}{4\lambda}.$$

Thus, type t 's hiring probability under pooling is

$$P_t^p = g_t P^p(C|\omega_G) + (1 - g_t) P^p(C|\omega_B) = \frac{1}{2} + \frac{2g_t - 1}{4\lambda},$$

so

$$P_L^p = \frac{1}{2} - \frac{1 - 2g_L}{4\lambda}, \quad P_H^p = 1 - P_L^p. \quad (29)$$

Separating equilibrium with both IC constraints binding. Under symmetry, $\gamma_L = \gamma_H =: \gamma$, so binding (26)–(25) implies

$$q_L(1 - q_L) = q_H(1 - q_H).$$

Since $q_L < q_H$, it follows that

$$q_L + q_H = 1.$$

Write

$$q_L = \frac{1}{2} - x, \quad q_H = \frac{1}{2} + x$$

for some $x > 0$. Binding IC-H yields

$$2x = \gamma \left(\frac{1}{4} - x^2 \right), \quad \gamma = \frac{1 - 2g_L}{2\lambda g_L(1 - g_L)},$$

with a unique positive root

$$x = \frac{-2 + \sqrt{4 + \gamma^2}}{2\gamma}.$$

Parts (i) and (ii). Using (29), the inequality $q_L < P_L^p$ is equivalent to

$$x > \frac{1 - 2g_L}{4\lambda}.$$

Let $\alpha := g_L(1 - g_L)$. Since $\gamma\alpha = (1 - 2g_L)/(2\lambda)$, the target becomes $x > \gamma\alpha/2$. Substituting the expression for x and simplifying gives

$$x > \frac{\gamma\alpha}{2} \iff 1 - 4\alpha > \gamma^2\alpha^2 \iff \lambda > \frac{1}{2}.$$

Thus, for $\lambda > 1/2$ it follows that $q_L < P_L^p$ and hence $q_H > P_H^p$, proving part (i).

Part (ii) follows from $q_L + q_H = 1 = P_L^p + P_H^p$.

Part (iii): manager gain. Substituting the interior optimizer Δ^{FOC} into (22) yields the maximized

value

$$V^*(\mu, q) = q(\mu - w) + \frac{q(1 - q)}{4\lambda}.$$

Under pooling, $(\mu, w, q) = (1/2, 1/2, 1/2)$, so the payoff is

$$\frac{1}{16\lambda}.$$

Under separation and symmetry, $w = 1/2$, $g_H = 1 - g_L$, and $q_H = 1 - q_L$. Writing $q_L = 1/2 - x$ and $q_H = 1/2 + x$, the manager's ex ante payoff is

$$\frac{1}{2}V^*(g_L, q_L) + \frac{1}{2}V^*(g_H, q_H) = x\left(\frac{1}{2} - g_L\right) + \frac{1/4 - x^2}{4\lambda}.$$

Subtracting $1/(16\lambda)$ yields

$$x\left(\frac{1}{2} - g_L\right) - \frac{x^2}{4\lambda} = x\left[\left(\frac{1}{2} - g_L\right) - \frac{x}{4\lambda}\right].$$

Let $\delta := 1/2 - g_L > 0$ and consider

$$f(x) := 2x - \gamma\left(\frac{1}{4} - x^2\right).$$

It follows that $f'(x) = 2 + 2\gamma x > 0$, so f is strictly increasing on $[0, 1/2)$. Also $f(0) = -\gamma/4 < 0$, and

$$f(\delta) = 2\delta - \gamma\left(\frac{1}{4} - \delta^2\right) = 2\delta - \gamma g_L(1 - g_L) = 2\delta - \frac{\delta}{\lambda} \geq 0$$

since $\lambda > 1/2$. Therefore, the unique root satisfies $x \leq \delta$, and hence

$$\delta - \frac{x}{4\lambda} \geq \delta\left(1 - \frac{1}{4\lambda}\right) > 0.$$

Thus, the payoff difference is strictly positive. □

Remark 2 (The knife-edge case $\lambda = 1/2$). At $\lambda = 1/2$, pooling implements perfect screening. However, in the separating problem, boundary optimality does not in general imply perfect screening.

Indeed, fix any $D \in (0, 1]$ and set

$$q_L = g_L D, \quad q_H = g_H D.$$

Under symmetry, $g_H = 1 - g_L$, and for each $t \in \{L, H\}$ it follows that

$$\Delta_t^{\max} = \min \left\{ \frac{q_t}{g_t}, \frac{1 - q_t}{1 - g_t} \right\} = D.$$

At $\lambda = 1/2$, the first-order spread is

$$\Delta^{FOC}(g_t, q_t) = \frac{q_t(1 - q_t)}{g_t(1 - g_t)} = D \frac{1 - g_t D}{1 - g_t} \geq D,$$

with strict inequality whenever $D < 1$. Hence, the quota-constrained optimum is at the boundary and implements

$$P(C|\omega_G, t) = D, \quad P(C|\omega_B, t) = 0.$$

The truthful hiring probabilities are then

$$P_L(C|L) = g_L D = q_L, \quad P_H(C|H) = g_H D = q_H,$$

and deviation payoffs satisfy

$$P_H(C|L) = q_L + (g_H - g_L)D = g_H D = q_H,$$

$$P_L(C|H) = q_H - (g_H - g_L)D = g_L D = q_L.$$

Thus, both IC constraints bind for every $D \in (0, 1]$. The special case $D = 1$ yields perfect screening and leaves candidate hiring probabilities unchanged relative to pooling, but for $D < 1$, candidates' welfare differs from pooling. For this reason, Proposition 9 focuses on the case $\lambda > 1/2$, where the relevant optimal hiring rules are interior.